

ITA0609-MACHINE LEARNING

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Question: 1

1. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

Answer

Introduction

A Machine Learning (ML) system can help doctors predict diseases such as heart disease or diabetes by analyzing patient information. The system uses patient data like age, blood pressure, glucose level, BMI, smoking habits, physical activity, and family history to predict whether a patient is at risk of developing a disease.

Step 1: Data Collection

Collect patient data from hospitals or healthcare databases.

Features used:

- Age
- Blood Pressure
- Glucose Level
- BMI
- Cholesterol
- Smoking Habit
- Physical Activity
- Family History

Target:

- Disease (Yes / No)

Step 2: Data Preprocessing

Before training the model, the data should be cleaned.

- Remove duplicate records.

- Handle missing values.
- Remove incorrect or noisy data.
- Convert categorical values into numerical values.
- Normalize or standardize numerical data.

This improves the quality of the dataset.

Step 3: Feature Selection

Select only the important features that affect disease prediction.

Important features include:

- Age
- Blood Pressure
- Glucose Level
- BMI
- Cholesterol
- Lifestyle Factors

Feature selection reduces unnecessary data and improves model performance.

Step 4: Choice of Algorithm

Algorithm: Random Forest Classifier

Reason for Selection:

- High prediction accuracy.
- Handles large datasets effectively.
- Works well with both numerical and categorical data.
- Reduces overfitting.
- Identifies important features.

Alternative Algorithms:

- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)

Among these, Random Forest is preferred because it provides better accuracy and reliability for disease prediction.

Step 5: Model Training

- Divide the dataset into Training Data (80%) and Testing Data (20%).
- Train the Random Forest model using the training dataset.
- Test the model using the testing dataset.
- Adjust model parameters to improve performance.

Step 6: Model Evaluation

Evaluate the model using performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

A higher accuracy indicates better disease prediction.

Working Process

Patient Data

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Data Collection

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Data Preprocessing

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Feature Selection

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Random Forest Model

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Model Training



Model Testing



Disease Prediction

(Heart Disease / Diabetes)

Advantages

- Early disease detection.
- Improves healthcare decisions.
- Fast and accurate prediction.
- Reduces manual diagnosis effort.
- Helps doctors provide better treatment.

Conclusion

A Random Forest-based Machine Learning system is suitable for predicting heart disease and diabetes because it provides high accuracy, reliability, and better generalization. Proper data preprocessing, feature selection, and model training ensure accurate disease prediction and support doctors in making better healthcare decisions.

Question 2

In the context of the Find-S algorithm, consider the following dataset used to determine whether a loan should be approved based on the attributes Employment Status, Salary Level, Credit Score, and Loan History.

Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Dataset

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Answer

Step 1: Initialize the Hypothesis

The Find-S algorithm starts with the most specific hypothesis.

$$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

Step 2: Process Example 1

Example 1:

(Employed, High, Good, Clean) → Yes

Updated hypothesis:

$$h_1 = \langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$$

Step 3: Process Example 2

Example 2:

(Employed, Medium, Good, Clean) → Yes

Compare with the previous hypothesis:

- Employment = Employed
- Salary changes from High to Medium, so replace it with ?
- Credit Score remains Good.
- Loan History remains Clean.

Updated hypothesis:

$$h_2 = \langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$$

Step 4: Process Example 3

Example 3:

(Unemployed, Low, Poor, Bad) → No

This is a negative example, so the Find-S algorithm ignores it.

The hypothesis remains:

$$h_3 = \langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$$

Step 5: Process Example 4

Example 4:

(Employed, Medium, Excellent, Clean) → Yes

Compare with the current hypothesis:

- Employment remains Employed.
- Salary already has ?, so it remains ?.
- Credit Score changes from Good to Excellent, so replace it with ?.
- Loan History remains Clean.

Updated hypothesis:

$$h_4 = \langle \text{Employed}, ?, ?, \text{Clean} \rangle$$

Final Hypothesis

$$h = \langle \text{Employed}, ?, ?, \text{Clean} \rangle$$

Conclusion

The most specific hypothesis consistent with all the positive training examples is:

$$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$$